Ownership Change and Business-Model Reorientation in Nursing Homes*

Joseph White† John Bowblis‡ Charles C. Moul§

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Abstract

Ownership changes in health care can reorganize what providers do, whom they serve, and how care is delivered. We study this process in U.S. nursing homes by identifying facility-level transfers of ownership control in CMS administrative data from 2017-2024 and tracing within-facility changes around those events. Ownership transfers are followed by substantial shifts in operating model: occupancy rises, Medicare share increases, Medicaid share falls, measured case mix increases, and average length of stay declines. Nursing labor also changes, with reductions in registered nurse and certified nursing assistant hours and little change in licensed practical nurse staffing. Quality responses are mixed. Some measures improve, including catheter use, urinary tract infections, hypnotic use, and short-stay pain, while preventive process measures such as influenza and pneumococcal vaccination deteriorate; several staffing-intensive resident outcomes show little average change. These patterns suggest that ownership transfers are not isolated governance changes applied to a stable care-production environment. Instead, they reveal broad business-model adjustment inside regulated health care providers, where changes in staffing and quality occur alongside simultaneous shifts in resident mix, payer composition, and throughout.

Keywords: nursing homes, ownership change, staffing, difference-in-differences

JEL Codes: I11, I18, J22, L22, L33

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1 Introduction

Ownership changes have become an increasingly common feature of the healthcare sector (footnote w/ citation?). Policymakers (Khan, 2024; Murphy and Scanlon, 2026; Oliveira and Blanco, 2024), academics (Stevenson et al., 2023; Ryskina et al., 2024; Gupta et al., 2024), and the public have expressed growing concern over the implications of these changes for patients and the quality of care. This concern is especially important in nursing homes as they serve an aging and medically vulnerable population who also face a wide range of frictions. Yet, existing literature gives limited or mixed results on how ownership changes affect patients, quality, and the facilities operating model. This is in part because ownership is difficult to measure and many studies only focus on specific forms of ownership, such as private equity, chain mergers, or for-profit acquisitions. This paper addresses that gap by introducing a new measure of verified substantive ownership changes and estimating their effects on both the production and quality of care in U.S. nursing homes.

This paper studies that question in U.S. nursing homes. We use CMS panel data from 2017 through 2024 to identify true transfers of economic control and estimate how staffing inputs and quality outcomes change following ownership changes. Nursing homes are a useful setting because prices are heavily shaped by public reimbursement, residents are medically vulnerable, and staffing is a central input into care production. If ownership changes affect care in this setting, the effects are likely to operate through internal production decisions: nurse staffing, staffing composition, task allocation, medication management, monitoring, documentation, and clinical routines.

The effect of ownership change is theoretically ambiguous because facilities may change owners for different reasons. Some transitions may reflect financial distress. A facility facing low occupancy, high labor costs, regulatory pressure, or declining profitability may be sold to an owner that expects to stabilize operations by reducing costs, tightening staffing, or standardizing management. In this case, ownership change may lead to input reductions, and whether those reductions harm residents depends on how close the facility was to the margin of adequate care. Other transitions may reflect succession or exit by incumbent owners. A small independent owner may sell because of retirement, limited access to capital, or limited ability to manage increasingly complex regulatory and administrative requirements. In these cases, a new owner may alter operations not because the facility is failing, but because the previous owner lacked the capacity or incentive to update practices.

A different set of ownership changes may reflect strategic acquisition or corporate restructuring. A chain may acquire a facility because it expects economies of scale in administration, purchasing, compliance, or labor management. A buyer may also acquire a facility because it believes existing operations can be reorganized without fundamentally changing the resident population or local market. Conversely, some ownership changes may be largely administrative or legal, producing little change in day-to-day care. These different motives imply different predictions: ownership change may generate quality-reducing cost containment, little operational change, or efficiency-enhancing reorganization. Which of these responses occurs is ultimately an empirical question.

Distinguishing among these possibilities requires examining both inputs and outcomes. Staffing reductions alone are difficult to interpret. They may reflect quality-reducing cost cutting, but they may also reflect the removal of inefficient routines or changes in the organization of care. Quality measures alone are also incomplete, because observed improvements may occur alongside reductions in labor inputs that matter for unmeasured aspects of resident welfare. The joint response of staffing and quality is therefore central. If ownership changes primarily reduce inputs in ways that harm residents, staffing declines should be accompanied by worsening in resident outcomes. If staffing falls while quality measures improve or remain stable, the pattern is more consistent with reorganization of production. In that case, ownership change may reveal that some facilities had previously operated inside their feasible production frontier: not because new owners have a superior technology, but because changes in control can disrupt routines, alter managerial attention, and change how labor is allocated across tasks.

Ownership changes are not random events, which creates an important empirical challenge. Facilities that change owners may differ from those that do not, and the conditions leading to a sale may also shape subsequent staffing and quality. Our analysis therefore focuses on within-facility changes around transfers of control rather than cross-sectional differences between facilities with different owners. In a staggered difference-in-differences event-study design, facilities are compared to themselves before and after ownership change, relative to contemporaneous changes among facilities that have not yet changed ownership or do not change ownership during the sample period. This design does not eliminate all concerns about endogenous timing, but it allows us to ask a more specific question: how do staffing and quality evolve when control of an existing facility changes hands?

This paper contributes to a growing literature on ownership, consolidation, and quality in health care. Prior work emphasizes that mergers and ownership changes can affect provider behavior through several channels, including market power, economies of scale, financial incentives, organizational restructuring, and changes in workforce composition. Much of the existing evidence focuses on hospital mergers, physician-practice consolidation, private equity acquisitions, or differences across ownership forms. These studies show that the effects of ownership and consolidation on quality are often mixed and context-dependent. Some mergers appear to reduce capacity without improving productivity or quality; others improve management or quality measures while worsening patient experience or leaving broader clinical outcomes unchanged. This evidence suggests that ownership changes should not be understood only as changes in market structure. They may also operate through internal production decisions.

Nursing homes are a particularly useful setting for studying these internal production channels. The sector is labor intensive, heavily regulated, and financed largely through public reimbursement. Because prices are constrained, staffing and care routines are natural margins of adjustment. Nursing home ownership has also been a persistent policy concern. Prior work finds that ownership form, chain affiliation, private equity ownership, and public reporting can affect staffing, quality, and resident outcomes. However, less is known about what happens when control of an existing nursing home changes hands while the facility itself remains in the same market. This distinction matters. A transfer of control does not necessarily alter the local set of providers, but it may change the decision-maker responsible for staffing levels, staffing mix, monitoring systems, medication management, documentation, and other care routines.

Nursing labor is the core input in nursing home care production, and staffing is one of the most consequential operation choices facilities make. We distinguish among registered nurses (RNs), licensed practical nurses (LPNs) and certified nursing assistants (CNAs). RNs include registered nurses and directors of nursing and play central supervisory and clinical roles, including assessing residents' conditions, developing and coordinating care plans, overseeing medication-related processes, and supervising other staff. LPNs provide routine clinical services and medication management tasks such as monitoring vital signs, while CNAs deliver the majority of hands-on daily care, assisting with bathing, dressing, toileting, feeding, and mobility. Because these staff types differ in training, scope of practice, and cost, ownership changes may affect not only total staffing hours but also the skill mix of care delivered. For example, an owner seeking quick cost containment may reduce total hours or substitute away from higher-cost licensed nurses, while an owner focused on compliance or clinical quality may increase RN oversight or adjust supervision structures.

Our results show that ownership changes are followed by sustained reductions in nursing labor inputs. RN hours and CNA hours decline, and total nursing hours fall. LPN hours remain largely unchanged. These staffing responses suggest that ownership transitions alter both the level and composition of labor used in care production. Our quality results complicate a simple cost-cutting interpretation. Labor saving mechanism measures, including catheter use and anti-anxiety or hypnotic medication use, improve after ownership change. Urinary tract infections also decline. By contrast, resident outcome measures such as pressure injuries, falls with major injury, weight loss, and decline in physical functioning show little evidence of broad worsening.

The paper makes three contributions. First, it contributes to the literature on health care ownership and consolidation by studying ownership change as a shock to internal production rather than only as a change in market structure. Second, it contributes to the nursing home literature by jointly examining staffing composition and measured quality outcomes following verified transfers of economic control. Third, it contributes to broader work on organizational change by showing that changes in governance can disrupt routines and alter production even when technology, location, regulation, and the local provider set remain largely fixed.

The following sections outline the rest of this paper. Section 2 provides background on the nursing home industry and ownership structures. Section 3 describes the data, ownership-change measurement, staffing and quality outcomes, and empirical strategy. Section 4 presents the results. Section 5 discusses interpretation and limitations. Section 6 concludes.

2 Institutional Background

3 Data and Methods

3.1 Data Sources

We utilize three main sources of data: Nursing Home Compare (NHC) Archive data, Healthcare Cost Report Information System (HCRIS), and Payroll Based Journal (PBJ) data. The NHC Archive contains monthly facility level information on ownership, chain affiliation, certification status, as well as quarterly reported quality measures. Second, we use HCRIS, also known as the Medicare Cost Reports, which contains annual cost reports submitted by every Medicare certified nursing home in the United States. The cost reports contain information about costs, revenue, financial data for each reporting facility, and dates of ownership changes. Lastly, we use the PBJ data to obtain staffing information. The PBJ data contains the resident census and the number of hours worked by various types of NH staff on a daily basis.

3.2 Sample selection

We utilize two samples in our study, the first is our staffing panel which is constructed at the facility-month level and the second is our quality metric panel which is constructed at the facility-quarter level. We construct both samples using the following criteria. First, we limit our samples to the 48 contiguous U.S. states from the period January 2017 to June 2024. Secondly, the NHC Archive data is matched to the HCRIS data on a unique nursing home provider number. Because HCRIS only contains information on freestanding nursing homes, our samples exclude hospital-based nursing homes. Hospital based nursing homes differ from freestanding facilities in their structure and financing. Because they are within hospitals, staffing policies and resource allocation may be jointly determined with hospital operations and respond to hospital shocks and regulations rather than nursing home specific incentives. We therefore exclude hospital based facilities to maintain a more homogeneous set of providers and to ensure ownership changes reflect changes in nursing home management rather than broader hospital system changes. Hospital based nursing homes make up about 12 percent of all nursing homes in the United States.

We also exclude facilities that at any time during our sample were government owned. In some states, facilities that are privately owned are able to change their legal ownership to government entities while still keeping control of a facility. This

3.3 Identifying Ownership Changes

No single source provides a complete record of substantive ownership changes in U.S. nursing homes. We therefore construct a facility-level dataset of ownership changes intended to capture transfers of substantive economic control, rather than purely legal or administrative changes. Existing studies of nursing home ownership changes and mergers generally rely on two primary sources: HCRIS and the NHC Archive. Each source contains useful information, but each is incomplete when used alone. We therefore use a two-step identification and verification process that combines reported ownership-changes in HCRIS with monthly ownership records in the NHC Archive.

HCRIS provides a useful starting point because it records whether a facility reported a change in ownership during the reporting period and, when applicable, the reported date of the change. These records identify ownership changes that facilities report through the cost-reporting process. However, HCRIS does not by itself establish whether a reported change reflects a substantive transfer of economic control. Some HCRIS-reported changes may capture legal restructuring, administrative reorganization, or changes in ownership where decision-making authority remains within the same ownership group. As a result, relying on HCRIS alone may overstate the number of ownership changes that are economically meaningful for the production of nursing care. Appendix Table **??** provides examples of HCRIS-reported ownership changes that are not classified as substantive transfers, and therefore why HCRIS-reported changes need verification.

The NHC Archive provides more granular ownership information. It reports monthly owner roles, owner types, owner names, and ownership percentages for each facility. This allows us to examine whether the individuals or entities controlling a facility change over time. However, the detail in the NHC Archive also introduces measurement challenges. Nursing home ownership structures are often complex and layered. For example, a limited liability company may directly own a nursing home, while the company itself is owned by other individuals, corporations, trusts, or partnership entities. Changes in listed owners, ownership layers, or reporting conventions can therefore appear as ownership changes even when economic control has not clearly changed. Programming comparisons of monthly NHC ownership records can consequently generate false positives if used without additional validation. Appendix Table **??** also provides examples in which NHC ownership records indicate large changes in listed owners or ownership shares but no corresponding HCRIS ownership-change flag is observed.

We combine the two sources to address these limitations. HCRIS provides reported ownership changes and dates, while the NHC Archive allows us to verify whether those events correspond to changes in the underlying ownership structure. In constructing the NHC Archive ownership-change measure, we identify ownership changes at the highest ownership level observed for a facility. Specifically, when multiple ownership layers are present, we prioritize indirect owners, then direct owners, then partnership-interest owners. This is intended to capture changes in economic control at the highest available ownership layer rather than changes in lower-level legal entities that may not correspond to a change in decision-making authority.

We define a substantive ownership change as a case in which a majority (i.e. greater than 50%) of ownership shares turns over between month $t-1$ and month t. This characterization is intended to capture transitions in which control of the facility changes hands, including cases where a new owner acquires most or all ownership shares. When ownership percentages are partially or completely missing, we normalize any reported percentages and, where necessary, assign equal weights across remaining owners. To reduce false positives from legal restructuring within the same family ownership group, we do not flag a change when at least 80 percent of the ownership stake remains within the same surname family.

Using these verified ownership changes, the final analytic sample includes nursing homes that either have no ownership change in either source or have exactly one ownership change in both sources, with the HCRIS and NHC ownership-change dates occurring within six months of one another. For treated facilities, we define the ownership-change month as the month in which HCRIS identifies the change. All dates are converted to calendar months, so changes occurring at the beginning or end of a month are treated identically. Appendix Table **??** summarizes the ownership-change classification process and the construction of the analytic ownership-change sample.

3.4 Nursing Staff Levels

The nursing staff types we use in our study are registered nurses (RNs), licensed practical nurses (LPNs), and certified nursing assistants (CNAs). RNs are the most highly trained nursing staff and are responsible for assessment, care planning, medication administration, and supervision of other nursing personnel. LPNs provide basic medical care under the supervision of RNs and physicians, including monitoring vital signs and administering certain treatments. CNAs provide the majority of direct daily care to residents, assisting with activities of daily living. Because these staff types differ in training, scope of practice, and compensation, changes in staffing composition may reflect different organizational responses to ownership changes. We therefore examine each staff category separately, as well as total nurse staffing.

Nursing staff levels are calculated from PBJ data, which report daily staffing hours and resident census. For each facility i, calendar month t, and staff type, we aggregate daily hours and resident days to construct hours per resident day (HPRD):

$$\text{HPRD}_{it} = \frac{\sum_{d \in t} H_{id}}{\sum_{d \in t} R_{id}}, \tag{1}$$

where H_{id} is the number of hours worked by the staff type in facility i on day d, and R_{id} is the number of resident days in facility i on day d. We construct separate HPRD measures for RNs, LPNs, and CNAs, and define total HPRD as the sum of the three type-specific measures. For quality specifications estimated at the facility-quarter level, we construct quarterly staffing measures analogously by summing hours and resident days over all days in the quarter before taking the ratio.

We also construct additional measures of data quality. First, we calculate a monthly PBJ coverage measure, defined as the share of days in the month with complete staffing records. This measure allows us to identify incomplete data in our sample and describe our data completeness. We use this measure for sample construction and descriptive purposes, excluding facilities and observations with low reporting coverage. Secondly, we track reporting gaps, defined as the number of months since a facility's previous PBJ observation, and use this variable in robustness checks that restrict the sample to facilities with more continuous reporting.

Finally, we exclude a small number of observations with implausible staffing levels: months in which RN and LPN HPRD are both zero, total HPRD is below 1.5 or above 12, or CNA HPRD exceeds 5.25. These observations are outliers that likely reflect reporting errors rather than genuine variation in staffing. These exclusion criteria align with the criteria outlined in the CMS NHC Technical Users Guide.

3.5 Quality Metrics

We measure quality using selected CMS quality measures from the NHC Archive. These measures are reported at the facility-quarter level and are constructed from resident assessments submitted through the Minimum Data Set (MDS). We focus on measures that are plausibly responsive to ownership-related changes in staffing, care routines, medication management, monitoring, and documentation. For each measure used in the analysis, lower values indicate better measured quality.

We group the selected measures into two categories. The first category consists of labor saving mechanism measures: catheter use, anti-psychotic medication use, and anti-anxiety or hypnotic medication use. Catheter use measures the percentage of long-stay residents with a catheter inserted and left in their bladder. Catheters may be clinically appropriate in some cases, such as urinary retention, selected wound-care situations, or end-of-life care, but unnecessary catheter use can increase infection risk and may reflect facility routines around toileting assistance and monitoring. Anti-psychotic medication use measures the percentage of long-stay residents who received an anti-psychotic medication and anti-anxiety or hypnotic medication use measures the percentage of long-stay residents who received an anti-anxiety or hypnotic medication. Both of these metrics capture medication-management practices among long-stay residents. These medications may be clinically appropriate for some residents, but their use can also reflect prescribing routines, behavioral-management practices, staff monitoring, and reliance on pharmacological approaches to resident care.

The second category consists of resident outcome measures: pressure injuries, falls with major injury, weight loss, decline in physical functioning, and urinary tract infections. These measures are more directly related to resident health and functioning and may be affected by staffing-intensive care processes. Pressure injuries may reflect repositioning, skin checks, hygiene, nutrition, and monitoring. Falls with major injury capture serious resident safety events that may be affected by supervision, mobility assistance, medication management, and environmental safety. Weight loss captures clinically meaningful unintended weight loss [1] and may reflect nutrition, feeding assistance, monitoring, or changes in resident condition. Decline in physical functioning captures worsening dependence in mobility or activities of daily living. Urinary tract infections may reflect resident frailty and clinical need, but also catheter use, hydration, toileting assistance, hygiene, monitoring, and documentation practices.

3.6 Other Controls

To account for facility characteristics that may be correlated with staffing, quality, and ownership changes, we included a number of control variables that are consistent with other studies (cite here). The following variables come from NHC and HCRIS and are used as controls in all specifications: Ownership type (government, non-profit, and for-profit), chain affiliation, total number of beds, occupancy rate, payer mix (percent of revenue coming from Medicare, and percent of revenue coming from Medicaid), and acuity. Our measure of acuity is case-mix provided by CMS in the NHC archive data. During our study period CMS altered the methodology used to construct case-mix indices. As a result, raw case-mix values are

[1] This does not include residents who are on a physician-prescribed weight-loss plan.

not directly comparable over time. To address this issue, we measure acuity using relative rankings rather than levels. Specifically, we construct case-mix quartiles within each state and calendar month.

3.7 Analytic Approach

Ownership changes occur at different dates across facilities during the study period. Because these changes are not randomly assigned, our empirical strategy does not compare facilities that change ownership to facilities that do not in levels. Instead, we exploit the timing of verified control-changing ownership transitions to compare changes within a facility before and after ownership change, relative to contemporaneous changes among facilities that have not yet experienced a change or never experience one during the study period. All specifications include facility fixed effects, time fixed effects, and observable time-varying facility characteristics.

This staggered timing raises a standard concern for two-way fixed effects designs. In staggered-adoption settings, conventional two-way fixed effects estimates can be sensitive to treatment-effect heterogeneity and comparisons involving already-treated units (Goodman-Bacon, 2021). We therefore emphasize event-study estimates using alternative comparison groups. This design is well suited to the institutional setting. Following an ownership change, the facility remains in the same location, operates under the same certification framework, and serves the same local market. As a result, any post-transition changes are likely to operate through internal production decisions, including staffing, routines, and management practices, rather than through a mechanical change in local market structure.

A complication in defining event time is that the recorded ownership change month may not coincide with the beginning of the transfer process. The ownership change date we observe in HCRIS is the effective date of the change, or the closing date. However, the sale process begins earlier, when the buyer and seller negotiate and sign a contract. Ideally, we would observe the contract signing date or the month in which transition planning begins.

Because these dates are not observed, the months immediately preceding the recorded closing date may already reflect operational responses to the pending transfer. During this pre-transfer period, incumbent managers or prospective owners may adjust staffing policies, scheduling, hiring, agency use, or other operating routines. Because staffing can adjust relatively quickly, these months may not represent untreated facility behavior. For this reason, the preferred staffing event-study specification excludes $\tau = -3, -2, -1$ and uses $\tau = -4$ as the reference period. For quality outcomes, which are measured quarterly, the timing concern is different. The ownership-change quarter, $\tau = 0$, may contain care, assessment, and documentation from both before and after the transfer. We therefore exclude $\tau = 0$ and use $\tau = -1$ as the reference period.

We estimate the following set of event-study regressions:

$$S_{im} = \sum_{\tau=-24, \tau \neq -4,-3,-2,-1}^{24} \beta_\tau \mathbf{1}[T_{im} = \tau] + X_{im}\gamma + \mu_i + \lambda_m + \varepsilon_{im}. \tag{2}$$

For staffing outcomes, S_{im} denotes RN, LPN, CNA, or total HPRD for facility i in month m. The variables $\mathbf{1}(T_{im} = \tau)$ are event-time indicators for whether facility i is τ months from ownership change. The coefficients β_τ trace monthly staffing dynamics relative to four months before the ownership change. The months $\tau = -3, -2, -1$ are excluded to remove the period most likely to be influenced by the pre-transfer adjustment window.

$$Q_{iq} = \sum_{\tau=-8, \tau \neq -1,0}^{8} \theta_\tau \mathbf{1}[T_{iq} = \tau] + X_{iq}\pi + \mu_i + \lambda_q + \nu_{iq}. \tag{3}$$

For quality outcomes, Q_{iq} denotes a selected CMS quality measure for facility i in quarter q. The variables $\mathbf{1}(T_{iq} = \tau)$ are event-time indicators for whether facility i is τ quarters from ownership change. The coefficients θ_τ trace quarterly quality dynamics relative to the quarter before ownership change. The ownership-change quarter, $\tau = 0$, is excluded because it may contain both pre- and post-transfer care, documentation, and reporting.

In both equations, X_{im} and X_{iq} denote vectors of facility controls including: ownership type, chain affiliation, beds, occupancy rate, payer mix, and state-period case-mix quartile indicators. Facility fixed effects μ_i absorb time-invariant differences across nursing homes, while month fixed effects λ_m and quarter fixed effects λ_q absorb common shocks over time. Standard errors are two-way clustered by facility and calendar period.

For quality outcomes, we estimate specifications both with and without RN, LPN, and CNA HPRD controls. Because staffing is itself affected by ownership change, the staffing-controlled specifications are not interpreted as preferred estimates of the total effect of ownership change on quality. Instead, they are used descriptively to assess whether the estimated quality responses are attenuated after conditioning on measured staffing inputs.

We also estimate average post-change specifications that replace the event-time indicators with a single post-ownership-change indicator:

$$Z_{ip} = \delta Post_{ip} + X_{ip}\gamma + \mu_i + \lambda_p + \varepsilon_{ip}. \tag{4}$$

In equation (4) $p = m$ for staffing outcomes and $p = q$ for quality outcomes. Z_{ip} denotes either a staffing outcome in facility-month im or a quality outcome in facility-quarter iq. The indicator $Post_{ip}$ equals one after the ownership change and zero otherwise. The coefficient δ summarizes the average post-ownership-change difference in the outcome, relative to the pre-change period, after accounting for facility fixed effects, time fixed effects, and time-varying controls. For staffing outcomes, the preferred post specifications are estimated after excluding the same immediate pre-transfer months used in the event-study specification. For quality outcomes, the preferred post specifications exclude the ownership-change quarter. These post specifications are used to summarize the average post-change difference and should be interpreted alongside the event-study estimates.

For staffing outcomes, we estimate event study and conventional difference-in-differences in both levels and logs. Log specifications allow coefficients to be interpreted as approximate percentage changes and reduce sensitivity to differences in baseline staffing levels across facilities. Log specifications are omitted when staffing is zero.

3.7.1 Parallel Trends

The identifying assumption of equations 2 and 3 is, absent an ownership change, treated facilities would have followed similar staffing and quality trends as not-yet-treated and untreated facilities. We assess this assumption using the pre-treatment coefficients in the event-study models and joint Wald tests of whether these coefficients are equal to zero.

For staffing outcomes, we compare pre-trend tests from the full pre-window to those from the preferred donut specification. The full pre-window includes the months immediately preceding the recorded ownership change and uses $\tau = -1$ as the reference period. The donut specification excludes $\tau \in \{-3, -2, -1\}$ and uses $\tau = -4$ as the reference period. Table 1 reports the results of these Wald tests. The improvement in the pre-trend tests after removing the immediate pre-transition months supports the interpretation that those months are influenced by the pre-transfer adjustment window.

Table 1: Joint Wald Tests of Pre-trends for Monthly Staffing

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: HPRD				
2 Year Full Pre-Window	93.04 (23)	39.36 (23)	36.25 (23)	48.73 (23)
	[0.0000]	[0.0181]	[0.0389]	[0.0013]
2 Year Window with Donut	27.52 (20)	34.10 (20)	11.36 (20)	23.90 (20)
	[0.1213]	[0.0254]	[0.9362]	[0.2466]
1 Year Window with Donut	11.23 (8)	6.39 (8)	1.43 (8)	3.73 (8)
	[0.1892]	[0.6039]	[0.9938]	[0.8806]
Panel B: Log(HPRD)				
2 Year Full Pre-Window	47.06 (23)	38.05 (23)	35.80 (23)	44.39 (23)
	[0.0022]	[0.0252]	[0.0433]	[0.0047]
2 Year Window with Donut	26.11 (20)	37.18 (20)	10.86 (20)	22.71 (20)
	[0.1622]	[0.0111]	[0.9498]	[0.3033]
1 Year Window with Donut	4.30 (8)	6.81 (8)	1.74 (8)	4.45 (8)
	[0.8294]	[0.5574]	[0.9879]	[0.8140]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
Tested windows and reference periods: 2 Year Full Pre-Window tests $\tau = -24$ to $\tau = -2$ with reference $\tau = -1$; 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).
Sample sizes (rows): 2 Year Full Pre-Window ($N = 630,420$); 2 Year Window with Donut ($N = 626,311$); 1 Year Window with Donut ($N = 626,311$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

For quality outcomes, we conduct analogous Wald tests using the quarterly event-study specification. The full pre-window tests whether the coefficients for quarters $\tau = -8$ through $\tau = -1$ are jointly equal to zero. Our preferred quality specification excludes the ownership-change quarter, $\tau = 0$, because this quarter may combine pre- and post-transfer care, assessment, and documentation. Table 2 reports the results of these Wald tests.

Table 2: Joint Wald Tests of Pre-Trends for Quarterly Quality Measures

Outcome	**2 Year Full Pre-Window**	**2 Year, with Donut**	**1 Year, with Donut**
Catheter	12.10 (7) [0.0972]	13.00 (7) [0.0721]	2.01 (3) [0.5707]
Antipsychotic	7.49 (7) [0.3793]	7.89 (7) [0.3425]	3.43 (3) [0.3301]
Hypnotics	3.29 (7) [0.8570]	3.33 (7) [0.8532]	0.53 (3) [0.9117]
Pressure injuries	12.76 (7) [0.0781]	12.62 (7) [0.0820]	1.77 (3) [0.6211]
Falls	4.13 (7) [0.7649]	3.88 (7) [0.7937]	0.24 (3) [0.9711]
Weight loss	8.09 (7) [0.3251]	7.82 (7) [0.3489]	3.60 (3) [0.3083]
ADL increase	13.10 (7) [0.0697]	12.68 (7) [0.0802]	2.21 (3) [0.5306]
UTI	5.98 (7) [0.5422]	5.95 (7) [0.5450]	0.97 (3) [0.8081]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.
The 2 Year Full Pre-Window tests $\tau = -8$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
The ownership-change-quarter-excluded specifications omit $\tau = 0$ because the quarter of ownership change may combine pre- and post-transfer care, assessment, and documentation.
The 1 Year Window tests $\tau = -4$ through $\tau = -1$, with $\tau = -1$ as the omitted reference period.
Sample sizes by outcome: 2 Year Full Pre-Window [Catheter=188,762; Antipsychotic=201,403; Hypnotics=150,667; Pressure injuries=140,715; Falls=197,246; Weight loss=198,690; ADL increase=200,578; UTI=196,502].
Sample sizes by outcome: 2 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
Sample sizes by outcome: 1 Year, with Donut [Catheter=187,542; Antipsychotic=200,132; Hypnotics=149,711; Pressure injuries=139,764; Falls=195,998; Weight loss=197,438; ADL increase=199,311; UTI=195,249].
All specifications include facility and quarter fixed effects and covariates: government + non_profit + chain + beds + occupancy_rate + pct_medicare + pct_medicaid + cm_q_state_2 + cm_q_state_3 + cm_q_state_4.

3.7.2 Additional Analyses

We further examine heterogeneity in the effects of ownership change by estimating separate models of staffing before and during the COVID-19 pandemic. Specifically, we estimate models for the pre-pandemic period, January 2017 through December 2019, and the pandemic period, April 2020 through June 2024. Labor market conditions for nursing staff changed substantially during the COVID-19 pandemic, including heightened turnover, changes in wages, and staffing shortages. Estimating models separately by period allows us to assess whether ownership changes had differential effects on staffing before and during this structural shift in the nursing labor market.

In this subgroup analysis, we estimate the same specifications as in equations 2 and 4, maintaining consistent controls, fixed effects, and clustering. This analysis sheds light on whether the staffing consequences of ownership changes differ across pre-pandemic and pandemic-era labor market conditions.

4 Results

4.1 Summary Statistics

Table 3 reports summary statistics for the monthly staffing sample. The final sample consists of approximately 7,806 unique skilled nursing facilities and 632,231 facility-month observations spanning January 2017 through June 2024. Of these facilities, 1589 experience an ownership change during the sample period.

Table 3: Summary Statistics

Variable	Mean	SD	N
Panel A: Staffing sample (facility–month)			
Staffing outcomes			
RN HPRD	0.432	0.309	554,045
LPN HPRD	0.802	0.326	554,045
CNA HPRD	2.118	0.530	554,045
Total HPRD	3.352	0.777	554,045
RN hours (monthly)	1,030	945	554,045
LPN hours (monthly)	2,031	1,382	554,045
CNA hours (monthly)	5,312	3,358	554,045
Total hours (monthly)	8,373	5,103	554,045
Facility characteristics			
Non-profit	0.191	0.393	554,045
Chain affiliation (baseline)	0.572	0.495	554,045
Beds	107.5	56.4	554,045
Occupancy rate (%)	77.2	15.3	554,045
% Medicare	13.5	11.4	554,045
% Medicaid	60.1	20.8	554,045
Average length of stay (days)	160.9	157.9	544,030
Acuity quartile 2	0.230	0.421	554,045
Acuity quartile 3	0.233	0.423	554,045
Acuity quartile 4	0.240	0.427	554,045
Panel B: Quality sample (facility–quarter)			
Quality measures			
Catheter use	1.610	2.135	173,838
Anti-psychotic medication use	14.447	9.371	172,519
Anti-anxiety/hypnotic medication use	20.238	10.319	173,036
Pressure injuries	7.922	5.310	122,915
Falls with major injury	3.341	2.831	176,296
Weight loss	6.300	4.886	171,896
Decline in physical functioning	14.809	8.797	165,126
Urinary tract infections	2.511	3.250	175,551
Staffing and facility characteristics			
RN HPRD	0.429	0.303	183,977
LPN HPRD	0.800	0.320	183,977
CNA HPRD	2.113	0.518	183,977
Total HPRD	3.342	0.762	183,977
Non-profit	0.190	0.393	182,894
Chain affiliation (baseline)	0.572	0.495	183,977
Beds	107.8	56.3	183,977
Occupancy rate (%)	77.3	15.2	180,689
% Medicare	13.9	12.7	181,263
% Medicaid	60.1	20.9	164,327
Acuity quartile 2	0.248	0.432	183,977
Acuity quartile 3	0.251	0.434	183,977
Acuity quartile 4	0.259	0.438	183,977

Notes: N is the number of non-missing observations for each variable. HPRD denotes hours per resident day; monthly hours are the HPRD numerator. Chain affiliation is measured at the start of each facility's panel and held fixed. Acuity quartiles are within-state, within-period case-mix quartiles, with quartile 1 omitted. For every quality measure, lower values indicate better measured quality.

Panel A: 6,867 facilities (1,413 with an ownership change), 2017/01–2024/06. Panel B: 6,854 facilities (1,413 with an ownership change), 2017Q1–2024Q2.

Both samples exclude hospital-based facilities, facilities outside the 48 contiguous states, and any facility

4.2 Staffing Responses to Ownership Change

Figure 1 presents event study estimates of the effect of ownership changes on nursing staffing outcomes (i.e., Equation 2). Following an ownership transition, staffing declines persist for many periods, with the largest and most precise effects occurring for registered nurses (RN), certified nursing assistants (CNA), and total nursing hours per resident day.

RN staffing declines immediately following ownership changes and remains far below pre-transition levels throughout the post-treatment period. CNA staffing exhibits a similar but slightly smaller decline, while total nursing hours mirror the combined reductions across staff types. In contrast, LPN staffing shows limited and statistically insignificant changes over time. These dynamic patterns indicate that ownership changes are followed by sustained reductions in labor inputs, rather than short-run staffing adjustments that quickly reverse.

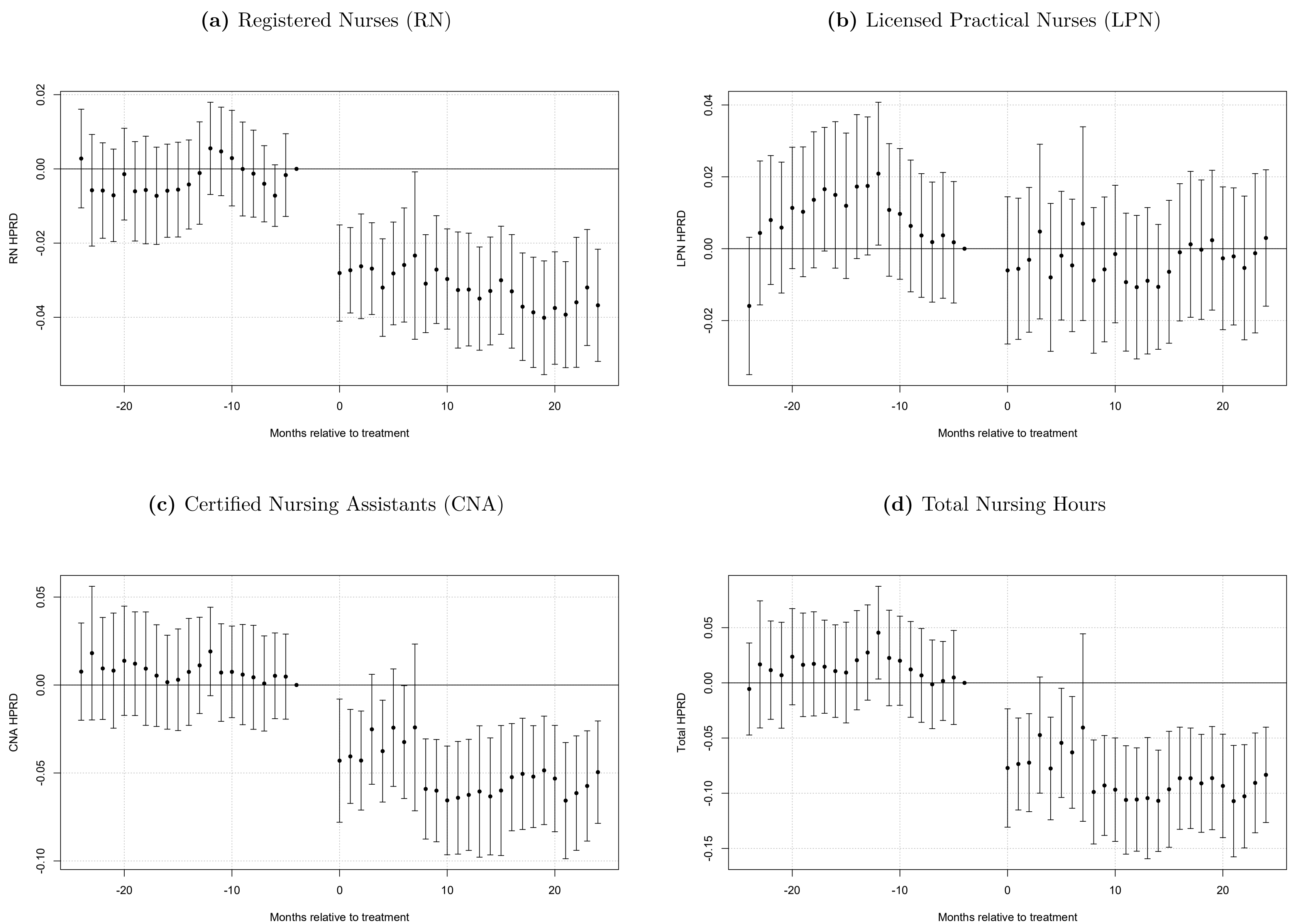


Figure 1: Dynamic Effects of Ownership Change on Nursing Staffing. Event study estimates of the effect of a change in ownership on nursing staff levels in hours per resident day (HPRD). Points show coefficient estimates relative to the reference period $\tau = -4$; vertical bars denote 95% confidence intervals. The three months prior to the ownership change ($\tau = -3, -2, -1$) are excluded. All specifications include control variables and facility and month fixed effects.

To summarize these dynamic effects with a single average treatment estimate, Table **??** reports two-way fixed effects estimates of the ownership change effect (i.e., Equation 4). Panel A reports results in levels, while Panel B reports logged results. Consistent with the event-study evidence, the average post-transition estimates indicate sustained reductions in nursing labor inputs. RN staffing declines by approximately 0.03 hours per resident day (roughly 9 percent), CNA staffing falls by about 0.05 hours (approximately 3 percent), and total staffing decreases by 0.08 hours per resident day (roughly 3 percent). LPN staffing shows no statistically significant change.

Table 4: Effect of Ownership Change on Nursing Staffing

Outcome	RN	LPN	CNA	Total
HPRD	-0.0368^{***}	-0.0039	-0.0532^{***}	-0.0940^{***}
	(0.0055)	(0.0062)	(0.0096)	(0.0134)
Log(HPRD)	-0.1174^{***}	-0.0022	-0.0270^{***}	-0.0293^{***}
	(0.0168)	(0.0095)	(0.0049)	(0.0041)
Hours	-80.0180^{***}	68.4756^{***}	64.8807^{*}	53.3383
	(15.3077)	(17.8010)	(34.9039)	(49.9070)
Log(hours)	-0.0767^{***}	0.0384^{***}	0.0133^{*}	0.0110^{*}
	(0.0173)	(0.0103)	(0.0068)	(0.0060)

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.
HPRD is hours per resident day; hours are the HPRD numerator, reported to test whether the HPRD decline is mechanically driven by the occupancy-rate increase in the denominator.
The anticipation window ($\tau = -3, -2, -1$) is excluded.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5: Effect of Ownership Change on Business-Model Outcomes

Outcome	Coefficient (SE)	Observations
Occupancy rate	2.6841***	550,330
	(0.3497)	
Spare capacity	−0.0263***	550,330
	(0.0035)	
Medicare share	1.6100***	550,330
	(0.2159)	
Medicaid share	−1.1710*	550,330
	(0.5952)	
Average length of stay	−13.3641***	540,346
	(4.3491)	
Case mix	0.0525***	457,842
	(0.0059)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Occupancy rate is residents as a share of available bed-days; spare capacity is unused certified beds as a share of certified beds. Payer shares are shares of patient days. The anticipation window ($\tau = -3, -2, -1$) is excluded.

Case mix is restricted to 2017/10–2023/12. CMS changed its case-mix methodology in October 2017 and January 2024; both breaks shift the level and the cross-sectional spread simultaneously across the whole sample, which facility and period fixed effects do not absorb.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2.1 Heterogeneity Before and After the Onset of COVID-19

Table 6 examines whether these effects differ before and after the COVID-19 pandemic. The estimated reductions are generally smaller during the pandemic period for CNA and total HPRD, while RN reductions remain negative and statistically significant. This pattern suggests that pandemic-era labor market constraints and heightened regulatory scrutiny may have limited some staffing adjustments after ownership change.

Table 6: TWFE Estimates of *post*: Pre- vs Post-pandemic Periods (Without anticipation)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
Panel A: Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.030^{***} (0.007)	-0.023^{**} (0.009)	-0.065^{***} (0.015)	-0.118^{***} (0.020)
Pandemic Period (2020/04 - 2024/06)	-0.020^{**} (0.007)	0.000 (0.008)	-0.045^{***} (0.016)	-0.064^{***} (0.021)
Panel B: Log Staffing Levels in HPRD				
Pre-Pandemic Period (2017/01 - 2019/12)	-0.091^{***} (0.026)	-0.017 (0.013)	-0.034^{***} (0.007)	-0.036^{***} (0.006)
Pandemic Period (2020/04 - 2024/06)	-0.075^{***} (0.022)	0.001 (0.011)	-0.022^{**} (0.008)	-0.021^{***} (0.006)

Notes: Each cell reports the coefficient on *post* with two-way clustered standard errors (by facility and month) in parentheses. Panel A reports levels (HPRD); Panel B reports logs (HPRD). Sample sizes: Row 1 ($N_{\text{levels}} = 222,399$; $N_{\text{logs}} = 222,399$), Row 2 ($N_{\text{levels}} = 312,471$; $N_{\text{logs}} = 312,471$).
All specifications include facility and month fixed effects and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.
Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.
Pre-pandemic 2017/01–2019/12; Pandemic 2020/04–2024/06.

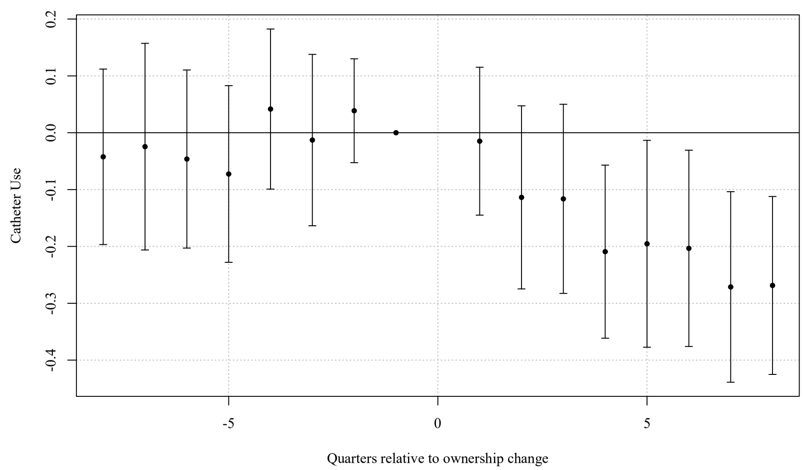
Together, the staffing results show that ownership changes are followed by sustained reductions in measured nursing labor inputs. The declines are concentrated among registered nurses, certified nursing assistants, and total nursing hours per resident day, while licensed practical nurse staffing remains comparatively stable. These patterns suggest that ownership changes are not merely legal or administrative events, but events of operational restructuring in which new owners adjust the mix and intensity of labor used in care production.

The welfare implications of these input reductions are ambiguous without examining quality outcomes. If staffing reductions primarily reflect harmful cost-cutting, one would expect worsening in resident outcomes, particularly measures that depend heavily on direct staff attention. If, instead, ownership changes disrupt old routines and move facilities closer to their feasible production frontier, reductions in labor inputs may occur alongside improvements in quality measures that are sensitive to managerial routines, monitoring, and care protocols.

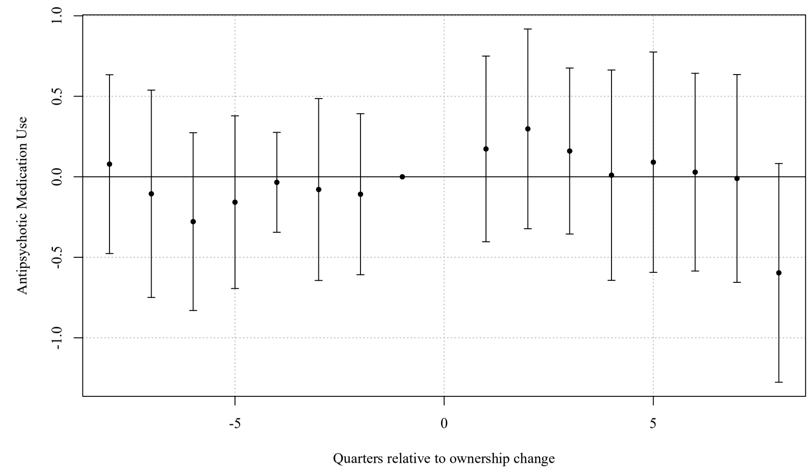
4.3 Quality Responses to Ownership Change

Figure 2 presents event-study estimates for routine-sensitive quality measures. These outcomes are intended to capture changes in care that may respond to changes in oversight, monitoring, documentation, and clinical routines. The estimates show whether ownership changes are followed by changes in quality measures that may be especially responsive to organizational re-optimization.

(a) Catheter Use



(b) Anti-Psychotic Medication Use



(c) Anti-Anxiety or Hypnotic Medication Use

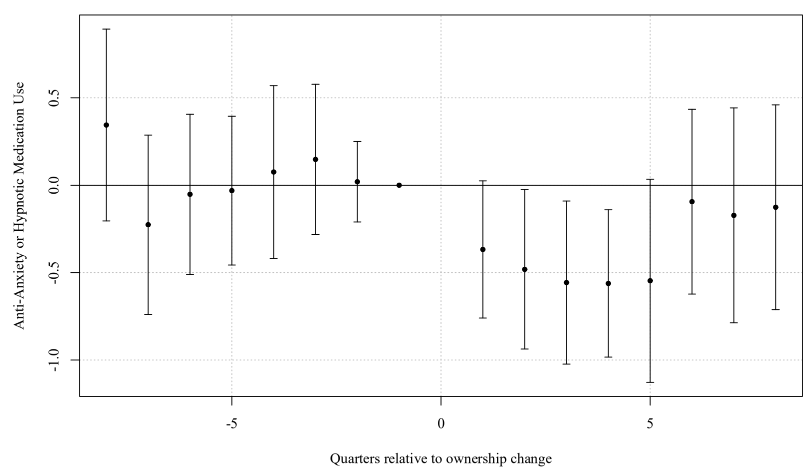


Figure 2: Dynamic Effects of Ownership Change on Labor Saving Mechanisms. Event-study estimates of the effect of a change in ownership on labor saving mechanism measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.

The results shown in figure 2 provide mixed but generally supportive evidence that ownership changes alter clinical and managerial routines. Catheter use declines following ownership change, and anti-anxiety or hypnotic medication use also falls relative to the reference period. Antipsychotic medication use declines minimally but is not statistically significant. Because lower values indicate better measured quality for all quality outcomes, these patterns suggest that ownership changes are not followed solely by reductions in measured inputs; some quality measures that capture labor saving mechanisms improve after control changes.

Figure 3 presents event-study estimates for staffing-intensive resident outcomes. These outcomes are more directly tied to hands-on care intensity. If ownership changes primarily reduced inputs in ways that harmed residents, these measures would be the most likely to worsen.

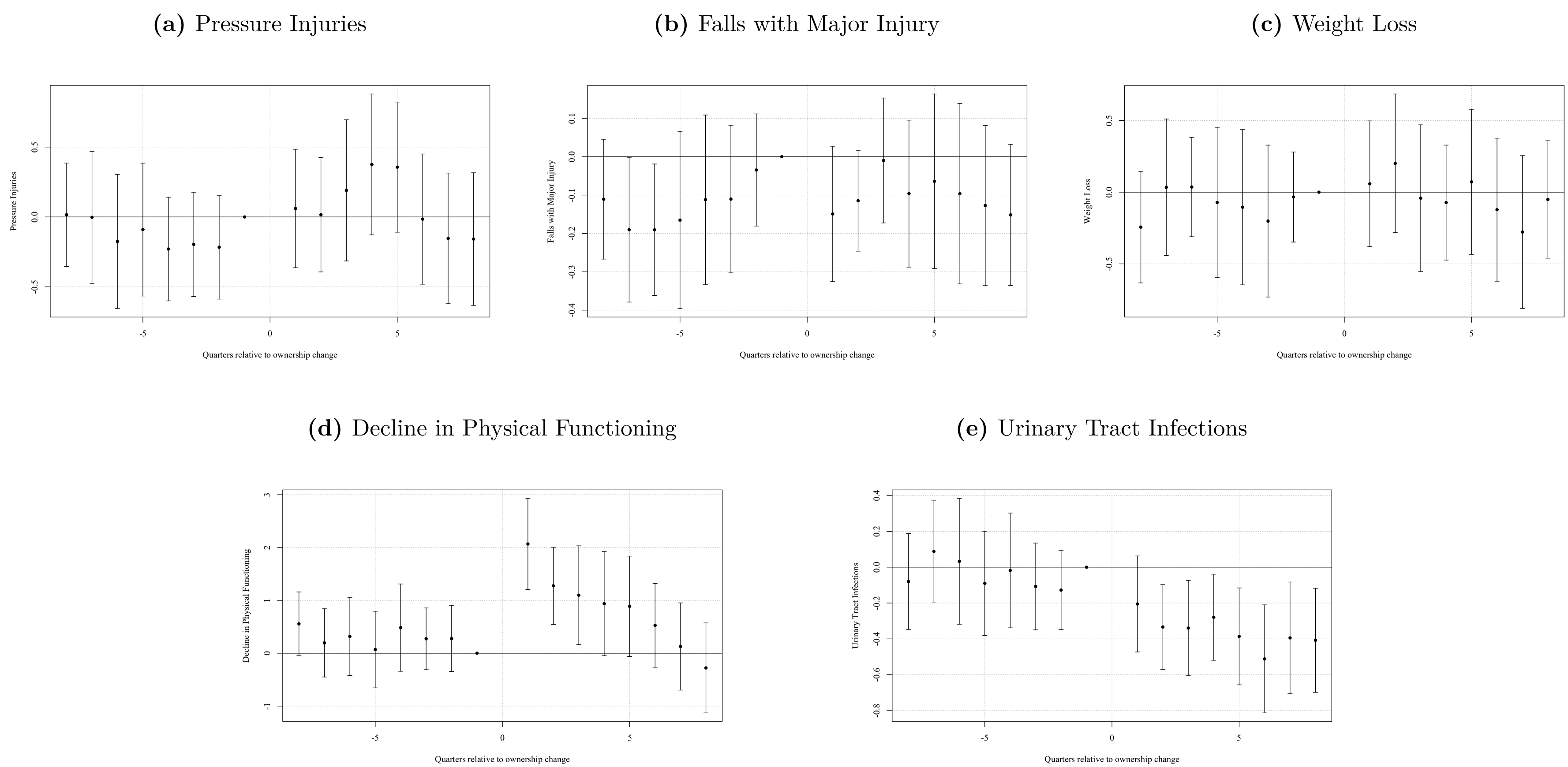


Figure 3: Dynamic Effects of Ownership Change on Resident Outcome Measures. Event-study estimates of the effect of a change in ownership on resident outcome measures. Points show coefficient estimates relative to the reference period $\tau = -1$; vertical bars denote 95% confidence intervals. The ownership-change quarter ($\tau = 0$) is excluded from the estimation sample. All specifications include control variables and facility and quarter fixed effects.

The resident outcome measures show less evidence of broad worsening. Pressure injuries, falls with major injury, weight loss, and decline in physical functioning do not change in a statistically significant manner. Urinary tract infections do show modest declines. Taken together, these results provide little evidence that staffing reductions, through ownership changes, translate into widespread worsening of the selected resident outcome measures over the observed window.

Table ?? summarizes the quality responses using average post-change TWFE specifications.

Table 7: Effect of Ownership Change on Quality Measures

Outcome	(1)	(2)	Observations
Panel A: Labor-saving mechanism measures			
Catheter use	-0.1695^{***}	-0.1711^{***}	172,671
	(0.0488)	(0.0488)	
Anti-psychotic medication use	-0.3362	-0.3543	171,358
	(0.2218)	(0.2231)	
Anti-anxiety/hypnotic medication use	-0.2321	-0.2118	171,854
	(0.2019)	(0.2027)	
Panel B: Resident outcome measures			
Pressure injuries	0.2586	0.2825	121,962
	(0.1762)	(0.1783)	
Falls with major injury	0.0125	0.0138	175,123
	(0.0636)	(0.0644)	
Weight loss	0.2042^{*}	0.2080^{*}	170,733
	(0.1040)	(0.1034)	
Decline in physical functioning	0.3240	0.3501	163,983
	(0.2695)	(0.2782)	
Urinary tract infections	-0.3279^{***}	-0.3181^{***}	174,381
	(0.0765)	(0.0767)	

Notes: Each cell reports the coefficient on *post*, with standard errors in parentheses. All specifications include facility and calendar-period fixed effects and control for the number of certified beds. Standard errors are two-way clustered by facility and calendar period. The sample excludes facilities government-owned at any point during the study period.

Column (1) is the baseline specification. Column (2) adds RN, LPN, and CNA hours per resident day as controls. Because staffing is itself affected by ownership change, column (2) is not a preferred estimate of the total effect; it is reported to show whether the quality response is attenuated after conditioning on measured staffing inputs.

For every measure, lower values indicate better measured quality. The transition quarter ($\tau = 0$) is excluded. Observations are reported for column (1); column (2) drops facility-quarters with missing staffing.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The estimates are similar with and without staffing controls. Because staffing is itself affected by ownership change, the staffing-controlled specifications are not interpreted as preferred total effects. Instead, their similarity to the baseline estimates suggests that the measured quality responses are not mechanically explained by contemporaneous changes in RN, LPN, and CNA hours per resident day.

4.4 Robustness

This section reports robustness checks for our staffing results, which are the main input-side estimates in our paper. These checks are designed to assess the validity of our identifying assumptions and the sensitivity of our results to alternative model choices. Across all exercises, the estimated effects of ownership change on staffing remain stable in sign, magnitude, and statistical significance. The supporting tables and figures are reported in the appendix.

4.4.1 Alternative Comparison Groups

Our main event-study specification is estimated using two-way fixed effects (TWFE) with staggered treatment timing. In this setting, TWFE can implicitly use already-treated facilities as controls for cohorts treated later, which may distort event-study estimates when treatment effects are dynamic or heterogeneous across cohorts. To assess sensitivity to this issue, we re-estimate the event study using a stacked difference-in-differences design.

In the stacked approach, we construct cohort-specific samples for each treatment month g. For a given cohort, treated facilities are those with an ownership change in month g, and the control group consists only of facilities that are never treated or not-yet-treated relative to g (i.e., with treatment dates after g). We restrict each cohort-specific sample to a symmetric event window around g and then stack these cohort datasets into a single analysis file. Estimation includes the same covariates as the baseline specification and fixed effects that absorb facility-level time-invariant heterogeneity and common calendar shocks; standard errors are clustered at the facility level.

Appendix Figures ?? through ?? show that the stacked event-study estimates closely track the baseline TWFE event-study results: the post-transition dynamics and the relative magnitudes across staffing types remain qualitatively unchanged. In addition, Appendix Table ?? reports joint Wald tests of pre-treatment coefficients in the stacked specification (under the donut design used in the baseline), which are consistent with flat pre-trends in the remaining pre-period. Together, these results suggest that the main findings are not driven by treated-as-control comparisons that can arise in TWFE under staggered adoption.

4.4.2 Event Window and "Donut"

As we have previously discussed, because of changes to staffing in anticipation of a formal ownership change, the assumptions of parallel trends is violated. As a result of this, in our baseline specifications we "donut" equation 2 and equation 4 by removing $\tau \in \{-3, -2, -1\}$. To test the validity of removing this set of pre-treatment coefficients, we estimate equations 2 and 4 with different "donut" sets and show that the estimated post-treatment effects remain largely unchanged. Row 6 of Table ?? reports estimates of equation 4 with a wider anticipatory period excluded from the sample. The results from row 6 show no differences in values or statistical significance. Similarly, row 7 reports estimates under a smaller anticipatory window that is excluded from the data, and estimates remain similar in magnitude and statistical significance.

4.4.3 Gap Size

Because staffing data is occasionally missing or reported with gaps, we examine whether our results are sensitive to the length of time between consecutive observations. Restricting the sample to facilities with smaller reporting gap shows that estimates are similar to those in the full sample, indicating that data gaps do not drive our findings. Row 2 of Table ?? reports our results for equation 4 with the restriction that facilities with a reporting gap of larger than 6 are excluded from our sample. Compared to the baseline (row 1) this result is identical across staff types and statistical significance. The exact same result is true for row 3 which reports estimates for equation 4 where facilities with a reporting gap of greater than 3 are removed. Row 4 and 5 report estimates where gap greater than 1 or equal to 0 respectively, are removed. Similarly, we see no differences in these results with respect to the baseline.

5 Discussion

This paper studies ownership changes in nursing homes as moments of organizational disruption in care production. Our results show that ownership transitions are followed by sustained reductions in measured nursing labor inputs, especially RN hours, CNA hours, and total nursing hours per resident day. At the same time, we find that some selected quality measures improve: catheter use and anti-anxiety or hypnotic medication use decline, and urinary tract infections also fall. Other resident outcome measures, including pressure injuries, falls with major injury, weight loss, and decline in activities of daily living, show little evidence of broad worsening. Taken together, these patterns suggest that ownership changes do more than alter the legal identity of the owner. They appear to change how facilities organize production.

The combination of lower staffing inputs and selected quality improvements is consistent with the idea that some facilities operate inside their feasible production frontier before undergoing ownership change. New owners do not necessarily possess a superior care technology. Instead, the ownership transition may disrupt old routines, alter managerial oversight, change medication-management practices, or reduce organizational slack. If facilities, under previous ownership, relied on inefficient routines, then new owners could reduce some inputs while maintaining or improving selected measured quality outcomes. This is consistent with the idea that ownership change may not be a mechanism for cost-containment, but rather a way for new owners to re-optimize production inputs.

5.1 Interpreting Staffing Reductions

The results of our staffing specifications indicate that both the level and mix of nurse staffing adjust following a change in ownership. The declines are concentrated among RNs and CNAs, while LPN staffing remains comparatively stable. In a regulated sector with limited short-run pricing flexibility, staffing is one of the most immediately adjustable cost-saving mechanisms available to new owners.

The relative stability of LPN staffing is informative because LPNs occupy an intermediate position in nursing home production. RNs perform higher-skill clinical and supervisory functions, while CNAs provide much of the hands-on daily care (Lin, 2014). LPNs can perform routine clinical tasks and help maintain day-to-day coverage, making them a potential margin of adjustment when facilities reduce RN and CNA hours. The absence of a comparable decline in LPN staffing is therefore consistent with compositional adjustment rather than uniform staffing cuts across all nurse types.

An important question that can be raised is whether the estimated staffing reductions are economically large. Benchmarking the estimates against baseline staffing levels helps provide context. In our sample, mean staffing levels are 0.427 RN HPRD, 0.796 LPN HPRD, 2.100 CNA HPRD, and 3.324 total HPRD. In our average post-treatment effect specification, ownership change reduces RN staffing by approximately 0.03 HPRD, CNA staffing by approximately 0.05 HPRD, and total staffing by approximately 0.08 HPRD. Relative to baseline staffing levels, these estimates imply declines of roughly 7 to 9 percent for RN staffing and about 3 percent for CNA and total staffing. Expressed in minutes, the total staffing reduction is approximately five fewer nursing minutes per resident-day. These changes may be modest and should not, by themselves, be seen as evidence of substantial welfare losses. However, staffing is a core input into care production, so even small reductions may matter in facilities operating close to staffing constraints or for residents requiring more intensive assistance.

5.2 Quality Responses and the Reorganization of Care

The results of our regressions on quality help distinguish between alternative interpretations of the staffing reductions. If ownership changes primarily reflected quality-reducing cost containment, we would expect staffing-intensive resident outcomes to become worse after the ownership change. The evidence does not show broad worsening in the selected resident outcome measures. Pressure injuries, falls with major injury, weight loss, and decline in physical functioning are not statistically distinguishable from zero in the average post-change specifications. Urinary tract infections decline.

At the same time, we find that labor saving mechanism measures improve. Catheter use falls, and anti-anxiety or hypnotic medication use declines after ownership change. Anti-psychotic medication use also declines moderately, though the average post-change estimate is not statistically significant. These measures may respond to changes in care routines, medication management, monitoring, documentation, and managerial oversight. Their improvement suggests that ownership changes may lead facilities to revise care processes rather than simply reduce labor inputs.

This pattern is consistent with incumbent nursing home owners operating their facilities inside their feasible production frontier. A facility can be inside its feasible production frontier if existing routines persist even when better combinations of inputs and outcomes are attainable. Ownership change may interrupt those routines by introducing new managers, new monitoring systems, different performance targets, or greater willingness to change staffing and care practices. Under this interpretation, lower RN and CNA hours do not necessarily imply a proportional decline in care quality. Instead, they may reflect a reorganization of production in which some inefficient practices are reduced while selected quality measures improve.

5.3 Heterogeneity and Constraints

The heterogeneity results of our additional analyses provide insight into the constraints facing new owners. Staffing reductions are smaller during the COVID-19 period for several outcomes, suggesting that pandemic-era labor market tightness, heightened scrutiny, and operational disruption may have limited owners' ability to adjust staffing. When labor markets are tight or regulatory attention is elevated, the feasible set of staffing reductions may narrow.

Differences by organizational structure may also matter. Chain-affiliated facilities may have access to internal labor markets, standardized management practices, and administrative support that allow them to absorb ownership changes with smaller staffing disruptions. Independent facilities may face tighter financial or managerial constraints, making staffing a more immediate adjustment margin. These patterns are consistent with the broader interpretation that ownership changes operate through internal production decisions and that the scope for adjustment depends on capacity and constraints.

5.4 Limitations

Several limitations should be considered when interpreting our results. First, despite our two-step approach to identifying ownership transitions, the timing of ownership change may still be measured with error. The recorded ownership change date may not capture earlier contract negotiations, transition planning, or operational changes that occur before the formal closing date. Our preferred staffing specification addresses this concern by excluding the immediate pre-transition months, and our preferred quality specification excludes the ownership-change quarter, but some timing error may remain.

Second, ownership changes are not randomly assigned. The empirical design helps address time-invariant facility differences and common shocks, but ownership changes may still coincide with unobserved facility-level shocks such as financial distress, management turnover, or local labor market changes. These shocks could affect both staffing and quality.

Third, the staffing and quality measures are imperfect. PBJ staffing data may contain reporting gaps or measurement error, although robustness checks suggest that reporting gaps do not drive the main staffing results. CMS quality measures are also limited. They capture selected process and resident outcome measures but do not fully measure resident welfare, resident experience, deficiencies, hospitalizations, mortality, or other aspects of quality.

Finally, our results should not be interpreted as proving that ownership changes improve welfare. The evidence shows that ownership changes reduce selected labor inputs while some quality measures improve and others do not worsen. This pattern is consistent with routine replacement and inside-frontier inefficiency, but it is also possible that unmeasured aspects of care worsen or that effects differ across resident types and facilities. Future work should examine whether these ownership changes affect broader outcomes and whether the observed input reductions are concentrated in facilities with greater pre-transition slack.

6 Conclusion

Ownership changes are common in U.S. nursing homes, yet their implications for care production remain poorly understood. This paper studies ownership transitions as moments of organizational disruption and reoptimization. Using CMS administrative panel data from 2017 through 2024, we identify ownership transitions that reflect changes in economic control and estimate their effects on staffing inputs and selected quality outcomes. We find that ownership changes are followed by sustained reductions in RN staffing, CNA staffing, and total nursing hours per resident day, while LPN staffing remains comparatively stable. These staffing responses indicate that ownership transitions alter the internal organization of care production rather than merely changing the legal identity of the facility owner.

Our results focused on quality complicate a simple cost-cutting interpretation. If ownership changes reduced staffing in ways that broadly harmed residents, we would expect worsening in staffing-intensive resident outcomes. Instead, labor saving mechanisms improve, including catheter use and anti-anxiety or hypnotic medication use, while urinary tract infections also decline. Other resident outcome measures, including pressure injuries, falls with major injury, weight loss, and decline in physical functioning, show little evidence of broad worsening over the observed post-transition window. These patterns suggest that input reductions after ownership change do not translate mechanically into measured quality declines.

Our findings are consistent with ownership transitions revealing inside-frontier inefficiency in nursing home production. New owners do not necessarily possess a superior care technology, but they may be more willing to revise old routines, alter staffing composition, reduce organizational slack, and change medication-management or monitoring practices. In this sense, ownership change may operate as a disruption that shifts facilities toward a more efficient input-quality combination. At the same time, the welfare implications remain ambiguous. Our analysis does not directly estimate a production frontier, and the quality measures used do not capture all dimensions of resident welfare. Unmeasured outcomes, resident experience, deficiencies, hospitalizations, and mortality may respond differently.

This paper contributes to the literature by shifting the focus from ownership changes as solely market-structure events to ownership changes as shocks to the internal organization of care. In nursing homes, where prices are heavily regulated and staffing is a central input, changes in control can reshape production even when the facility remains in the same market and continues serving the same local population. Future work should examine which types of ownership changes generate efficiency-enhancing reorganization, which produce quality-reducing cost containment, and whether the observed input reductions are concentrated in facilities with greater pre-transition organizational slack.

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Tables

Table 8: Examples of Discordant Ownership Records Across HCRIS and NHC

Example	Classification	Period	NHC owner names and shares
Panel A: HCRIS-reported ownership changes not verified as substantive transfers in NHC records			
A	HCRIS only	$t = 1$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (50%); The Gail Miller GST (50%)
A	HCRIS only	$t = 2$	Karen Gail Miller Irrev. Tr. FBO Zane Miller (20%); The Gail Miller GST (20%); Karen Miller-related trust entities (60%)
B	HCRIS only	$t = 1$	Vetter Eldora (50%); Vetter Jack (50%)
B	HCRIS only	$t = 2$	Vetter Senior Living (100%)
C	HCRIS only	$t = 1$	Abri Health Services (50%); The Senior Care Liquidation Co., Alan Halperin Trustee (50%)
C	HCRIS only	$t = 2$	Abri Care (100%)
D	HCRIS only	$t = 1$	Concordia Lutheran Ministries of Pittsburgh (100%)
D	HCRIS only	$t = 2$	Concordia Care Network (100%)
Panel B: NHC-identified ownership changes not reported as ownership changes in HCRIS			
E	NHC only	$t = 1$	Destefane Richard (100%)
E	NHC only	$t = 2$	Richard J. Destefane Revocable Living Trust (100%)
F	NHC only	$t = 1$	D'Angelo Cara (25%); Durham Deborah (25%); Horace D'Angelo Jr. Operating Entities Dated 01/01/2012 (25%); Tackett Kimberly (25%)
F	NHC only	$t = 2$	Benoit (9.1%); Calumet South (9.1%); Drake Louis Enterprise (9.1%); Fairhome Tr. UAD 12312012 (9.1%); Fairhome UAD 12312012 (9.1%); GZLT (9.1%); Hartman David (9.1%); Hartman Mark (9.1%); Krupp Ari (9.1%); Senderowicz Yossi (9.1%); Willow Delta (9.1%)
G	NHC only	$t = 1$	CMC II Investors (33.3%); Conte Joseph (33.3%); FC LV Holdco (33.3%)
G	NHC only	$t = 2$	Qazi Mohammad (100%)
H	NHC only	$t = 1$	Berkshire Healthcare Systems (67%); Integritus Healthcare (25%); Berkshire Healthcare System (3%)
H	NHC only	$t = 2$	Integritus Healthcare Management Services (100%)
I	NHC only	$t = 1$	DES C 2009 GRAT (100%)
I	NHC only	$t = 2$	DES 2009 Family (100%)

Notes: This table presents examples of discordant ownership-change classifications across HCRIS and the NHC Archive in a panel-style format. Panel A reports cases in which HCRIS reports an ownership change, but the NHC ownership records do not show a clear transfer to a new unrelated controlling owner. Panel B reports cases in which NHC ownership records show large changes in listed owners or ownership shares, but no corresponding HCRIS ownership-change flag is observed. Owner names and shares are taken from the NHC Archive. The labels $t = 1$ and $t = 2$ refer to the ownership records immediately before and after the relevant ownership-record change.

Table 9: Ownership-Change Classification and Sample Construction

Panel A. Cross-classification of ownership-change indicators

	NHC ownership records		
HCRIS ownership-change status	0	1	2+
0	XXX	XXX	XXX
1	XXX	XXX	XXX
2+ changes	XXX	XXX	XXX

Panel B. Construction of analytic ownership-change sample

Sample restriction / classification step	Facilities
Matched NHC–HCRIS facilities after initial ownership-record cleaning	XXX
Exclude hospital-based nursing homes	XXX
Exclude facilities outside the 48 contiguous U.S. states	XXX
Exclude facilities with unresolved or missing provider identifiers	XXX
Exclude facilities with multiple ownership changes in either source	XXX
Keep facilities with no ownership change in either source	XXX
Keep facilities with exactly one ownership change in both sources	XXX
Restrict matched ownership-change dates to within six months of one another	XXX
Final analytic ownership-change sample	XXX

Notes: Panel A reports the cross-classification of facility-level ownership-change indicators across HCRIS and the NHC Archive after common ownership-record cleaning and pruning. NHC ownership changes are classified using monthly ownership records and are intended to capture substantive transfers of economic control. Panel B reports the stepwise construction of the analytic ownership-change sample. The final sample includes facilities with no ownership change in either source and facilities with exactly one ownership change in both sources, where the HCRIS and NHC ownership-change dates occur within six months of one another. Counts are at the facility level.

Table 10: TWFE DiD Estimates of *post* on Staffing Outcomes (Stacked Sample, Baseline Donut)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
HPRD	-0.024^{***}	-0.009	-0.051^{***}	-0.084^{***}
	(0.004)	(0.005)	(0.009)	(0.012)
Log(HPRD)	-0.070^{***}	-0.005	-0.027^{***}	-0.027^{***}
	(0.014)	(0.008)	(0.004)	(0.004)

Notes: Each cell reports the coefficient on *post* with standard errors in parentheses. The stacked sample includes never-treated and not-yet-treated controls for each cohort; the donut excludes $\tau \in \{-3, -2, -1\}$. Sample sizes: $N_{\text{HPRD}} = 24,197,338$; $N_{\text{Log}} = 23,819,333$.

Specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators.

Standard errors are clustered two ways by facility and calendar month.

Statistical significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 11: TWFE estimates of *post* (without anticipation).

	N	Outcomes: RN	LPN	CNA	Total
Panel A: Staffing Levels in HPPD					
(1) Baseline	628,107	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(2) Sample excludes *gap* > 6	627,923	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(3) Sample excludes *gap* > 3	627,626	-0.032^{***}	-0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(4) Sample excludes *gap* > 1	623,585	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(5) Sample excludes *gap* > 0	623,585	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.032^{***}	0.000	-0.055^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.032^{***}	0.000	-0.054^{***}	-0.086^{***}
		(0.005)	(0.006)	(0.009)	(0.013)
Panel B: Log Staffing Levels in HPPD					
(1) Baseline	628,107	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(2) Sample excludes *gap* > 6	627,923	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(3) Sample excludes *gap* > 3	627,626	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(4) Sample excludes *gap* > 1	623,585	-0.092^{***}	0.004	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(5) Sample excludes *gap* > 0	623,585	-0.092^{***}	0.004	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(6) Drop $t \in \{-4, -3, -2, -1\}$	626,729	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)
(7) Drop $t \in \{-2, -1\}$ only	628,107	-0.092^{***}	0.003	-0.028^{***}	-0.026^{***}
		(0.016)	(0.009)	(0.005)	(0.004)

Notes: Each cell reports the *post* coefficient with two-way clustered standard errors at the facility and month levels. Panel A uses staffing levels (HPPD); Panel B uses logs of HPPD.
All specifications include facility and month fixed effects and controls for ownership (government, non-profit, chain), beds, occupancy rate, payer mix (Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 12: Joint Wald Tests of Pre-Trends: Stacked Event Study (Levels)

	Outcomes			
	RN	**LPN**	**CNA**	**Total**
2 Year Window with Donut	34.45 (20)	20.65 (20)	20.06 (20)	25.86 (20)
	[0.0233]	[0.4178]	[0.4539]	[0.1706]
1 Year Window with Donut	8.62 (8)	11.90 (8)	5.48 (8)	11.83 (8)
	[0.3752]	[0.1558]	[0.7048]	[0.1588]

Notes: Each cell reports the Wald χ^2 statistic for the joint null that all pre-treatment event-time coefficients equal zero, followed by degrees of freedom in parentheses and the p-value in brackets.

Tested windows and reference periods: 2 Year Window with Donut tests $\tau = -24$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$); 1 Year Window with Donut tests $\tau = -12$ to $\tau = -5$ with reference $\tau = -4$ (dropping $\tau = -3, -2, -1$).

Sample sizes (stacked rows): 2 Year Window with Donut ($N = 24, 197, 338$); 1 Year Window with Donut ($N = 13, 268, 799$).

Stacked design: each treated cohort is compared only to never-treated and not-yet-treated facilities within its own event window; the donut excludes $\tau = -3, -2, -1$ (physically dropped from the estimation sample for treated units).

All specifications include facility-by-cohort fixed effects (stack_id), calendar-month fixed effects, cohort fixed effects, and covariates: *government*, *non-profit*, *chain*, *beds*, *occupancy rate*, *percent Medicare*, *percent Medicaid*, and state case-mix quartile indicators. Standard errors are clustered by facility.

Figures

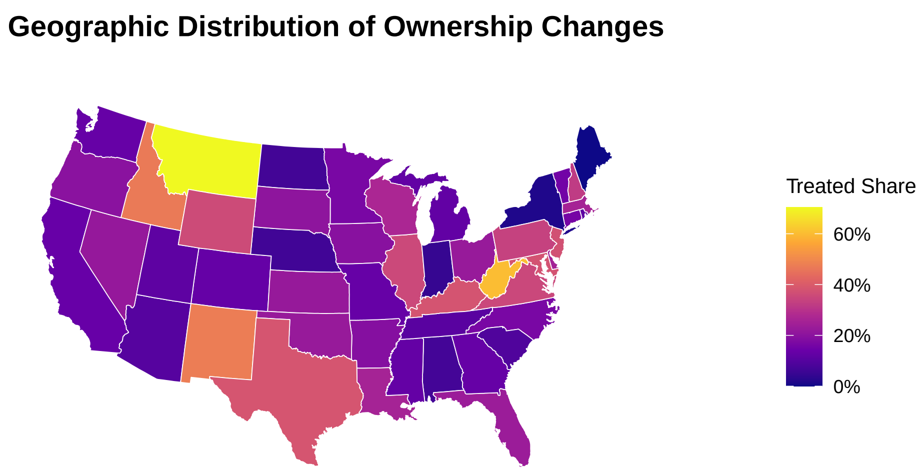

Figure 4: Geographic Distribution of Ownership Changes. The figure reports the share of nursing homes in each state that experienced an ownership change during the study period.

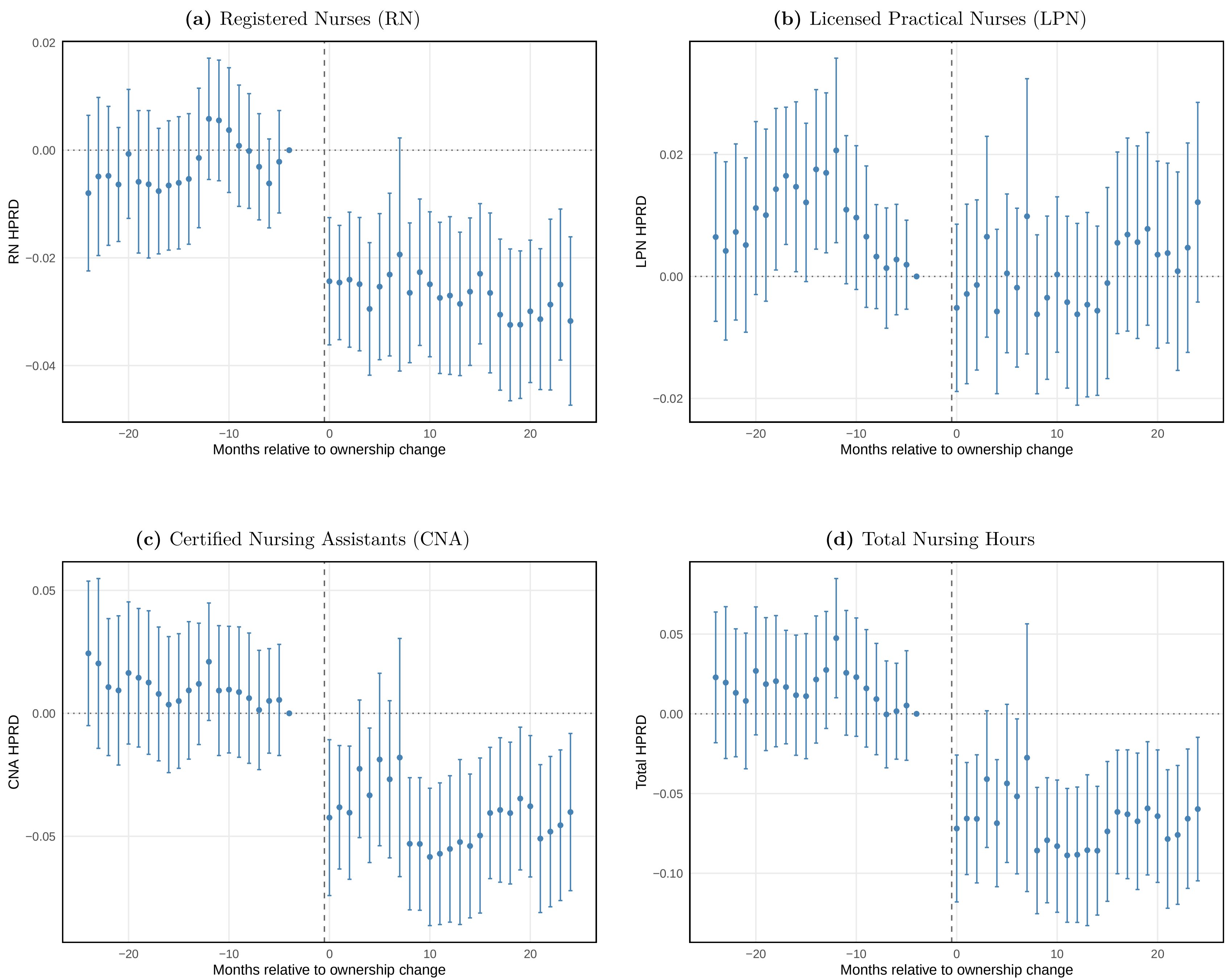

Figure 5: Stacked Event-Study Estimates of Ownership Change on Nursing Staffing. The stacked design compares each treated cohort only to never-treated and not-yet-treated facilities within the event window, preventing already-treated facilities from serving as controls. The donut excludes $\tau \in \{-3, -2, -1\}$; the reference period is $\tau = -4$.